

# Panel: Science of Design

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## ABSTRACT

In this session, leading scholars of design will share perspectives on a *science of design* for software and software-intensive systems. The session will include presentations by the panelists and dialog among the panelists and with audience members.

## Categories and Subject Descriptors

D.2.0 [Software Engineering]: General.

**General Terms:** Design.

## 1. OVERVIEW

The build-out of software-intensive systems that started about fifty years ago has transformed the world's economies, institutions, and cultures. Rapid growth and change will surely continue. The frantic pace of discovery and innovation to date has, in some ways, hindered reflection on what we've enabled, what kind of software-intensive society we wish to promote for our descendants, and what new knowledge and perspectives we will need in order to design systems in ways to foster the creation and preservation of a healthy, prosperous, open society.

This year's panel features leading scholars of software-intensive system *design*—broadly construed—who will present perspectives on prospects, problems and priorities for a *science of design* focused on software-intensive systems. Issues include the goals of such a science; the phenomena that are its subject; the form and testing of its theories; major open problems; the relationship of such a science to current research in the field; short- and longer-term challenges in developing and teaching an effective theory and practice of design; and intellectual and societal benefits expected to emerge from research in this area.

## 2. PANELISTS

The three panelists are Mary Shaw, Carliss Baldwin, and Michael Jackson. Kevin Sullivan will moderate.

**Mary Shaw** is the Alan J. Perlis Professor of Computer Science and Co-Director of the Sloan Software Industry Center at Carnegie Mellon University. She has been a member of this faculty since completing the Ph.D. degree at Carnegie-Mellon in 1972. Her research interests in computer science lie primarily in the areas of software engineering, particularly value-driven

software design, appropriate dependability, and software architecture.

**Carliss Y. Baldwin** is the William L. White Professor of Business Administration at the Harvard Business School. A specialist in corporate finance and real options theory, Baldwin received a bachelor's degree in economics from MIT in 1972, and MBA and DBA degrees from Harvard Business School in 1974 and 1977 respectively. She is active in Harvard's Ph.D. program in Information, Technology and Management, jointly sponsored by the Business School and the Division of Engineering and Applied Science. With Kim B. Clark, she is involved in a multi-year project to study the process of design and its impact on the structure of the computer industry. She and Clark have authored *Design Rules: The Power of Modularity*, the first of a projected two volumes on this topic. Volume 2, in progress, will focus on *The Institutions of Innovation*.

**Michael Jackson** has worked in software development since 1961. He developed the JSP program design method, chosen as the standard for UK Government software development, and he played the leading role in developing the JSD method of system analysis, specification and design. Since 1990 he has worked as an independent consultant. From 1990 to 2002, in conjunction with Dr Pamela Zave, he also worked as a part-time researcher at AT&T Bell Laboratories (now AT&T Research Laboratories) in New Jersey, USA. This work focused on the feature interaction problem, and on principles and techniques for specifying telecommunications and services, and resulted in the DFC (Distributed Feature Composition) virtual architecture for telecommunications services. His most recent work has focused on the analysis and structure of software development problems, using an approach based on the idea of problem frames. Problem frames can be regarded as patterns, in the problem space rather than in the solution space, with a major emphasis on the composition of identified sub-problems.

**Kevin Sullivan** is associate professor and VEF Faculty Fellow in Computer Science at the University of Virginia. He received his Ph.D. in Computer Science and Engineering in 1994 from the University of Washington. His interests are in the structures, properties, and value of design, with a particular emphasis on issues of modularity and integration in design and dependability as a user-perceived property.

## 3. ACKNOWLEDGEMENTS

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